Replication of Huggett & Ventura (1999)

"On the Distributional Effects of Social Security Reform"

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October 23, 2026

1 Introduction

Huggett & Ventura (1999, hereafter HV1999) ask who wins and who loses from replacing the U.S. social security system with the two-tier scheme proposed by the Bipartisan Commission on Entitlement and Tax Reform (the "Boskin proposal"): a defined-contribution annuity proportional to the social security taxes an agent has paid, together with a floor benefit $\underline{b}$ paid to any retiree whose annuity falls below it. The paper's central finding is distributional rather than aggregate. Aggregates barely move across the reforms, but the welfare consequences are sharply graded by ability, and at positive floor levels they are *U-shaped*: low- and high-ability agents gain while agents of median ability lose.

This document replicates the paper with the VFI Toolkit. Everything in HV1999 that reports model output is covered: Tables III–VI and Figures 1–6, under both of the paper's two wage specifications. Section 2 states the model, Section 3 the calibration and the data inputs, and Section 4 the equilibrium conditions — in particular how the floor benefit is financed, which the paper specifies in two places that have to be read together. Section 5 presents the results and Section 6 the differences from the paper that remain. Everything specific to the numerical implementation is gathered in Appendix A.

The replication is close on aggregates and on the qualitative distributional findings. K/Y is within 0.05 of HV1999 in every row of Tables III and IV, the interest rate within 0.2 percentage points, and the income Gini matches to the two digits the paper reports. The U-shape in the compensating variations replicates, as does the paper's argument about why retaining the medical transfer mitigates the welfare losses. Two quantities are off: the aggregate labour input L and the fraction of time worked each sit about 8 and 11 percent above the paper in *both* specifications and at every floor level, and the welfare aggregates of Tables V and VI run roughly twice as negative as HV1999's.

2 Model

Agents live at most $J = 81$ periods, entering at real-life age 20 and facing age-dependent survival probabilities s_j. There is no annuity market, so the assets of the deceased are accidental bequests. All variables below are detrended by the technology level, which is where the $(1 + g)$ factors come from. Parameter values are collected in Table 1; all are HV1999's, taken from the paper's Table I.

Parameter	Description	Value
J	Maximum lifespan (periods)	81
J_r	Retirement age (period)	46
β	Discount factor	1.011
σ	Curvature of the utility aggregator	2.0
ν	Consumption share in the aggregator	0.33
α	Capital share	0.36
δ	Depreciation rate	0.06
A	Total factor productivity	0.895944
n	Population growth rate	0.012
g	Technological growth rate	0.021
G/Y	Government consumption share	0.195
b/Y	Common medical/hospital transfer	0.1242

Table 1: Parameter values, all from Huggett & Ventura (1999). The wage-process parameters are specification-specific and are given in Section 3.

An agent of age j holds assets a, average past earnings $\bar{e}$ and productivity shock z, and chooses hours $l \in [0, 1)$ and next-period assets a'. Writing the problem recursively, with $x = (a, \bar{e}, z)$,

$$V_j(a, \bar{e}, z) = \max_{l \in [0,1),\; a' \geq 0} \left\{ \frac{\left(c^{\nu}(1-l)^{1-\nu}\right)^{1-\sigma}}{1-\sigma} \;+\; \beta\, s_j\, E\big[\, V_{j+1}(a', \bar{e}', z') \,\big|\, z \,\big] \right\}$$

subject to the budget constraint

$$c + (1+g)a' = a\big(1 + r(1-\tau)\big) + (1 - \tau - \theta_j)\, e(z, j)\, w\, l + b(x, j),$$

the law of motion for $\bar{e}$ given below, and the terminal condition $V_{J+1} \equiv 0$, where $\theta_j = \theta$ while working and 0 in retirement, $e(z, j) = \exp(z)\kappa_j$ is the agent's labour endowment in efficiency units, and $b(x, j)$ is the social security benefit. Retirement is mandatory at J_r. Production is Cobb–Douglas and government consumption is a fixed share of output.

Average past earnings evolve as HV1999 equation (7), which is

$$\bar{e}' = \frac{\bar{e}\,(j-1) + \min\{\, e(z, j)\, l,\; e_{\max} \,\}}{j} \qquad (j < J_r), \qquad \bar{e}' = \bar{e} \quad (j \geq J_r),$$

where $e_{\max}$ is the cap on creditable earnings. Writing the state instead in units of mean working-age earnings, so that the bend points and the cap sit at fixed grid values regardless of the equilibrium wage, this becomes

$$\bar{e}' = \frac{\bar{e}\,(j-1) + \min\big(e(z, j)\, l / \bar{y}^{\,\mathrm{mean}},\, 2.47\big)}{j} \qquad (j < J_r), \qquad \bar{e}' = \bar{e} \quad (j \geq J_r).$$

Here $\bar{y}^{\,\mathrm{mean}}$ is average earnings per working-age person measured in efficiency units, $E\big[\exp(z)\,\kappa_j\, l \mid j < J_r\big]$. It is distinct from the age profile κ_j: it depends on the equilibrium allocation, through hours and the distribution of z, and so is carried as an equilibrium unknown (Section 4). The cap of 2.47 and the bend points at 0.20 and 1.24 are then fixed numbers on the $\bar{e}$ grid, which is the point of writing the state this way.

Under the Boskin proposal $\bar{e}$ is replaced by a_2, the accumulated value of social security taxes paid, which follows HV1999 equation (10),

$$(1+g)\, a_2' = \big(a_2 + \theta\, e(z, j)\, w\, l\big)(1 + r) \qquad (j < J_r).$$

Benefits

Under the U.S. system (HV1999 equation 15) benefits are zero before J_r and thereafter

$$b(x,j) = b + \frac{w\,\bar{y}^{\,\text{mean}}\,\text{PIA}(\bar{e})}{(1+g)^{j-J_r}},$$

where b is a common medical/hospital transfer calibrated to $0.1242\,Y$ and PIA is the 1990–94 U.S. primary insurance amount schedule: 90% of the first 0.20, 32% to the second bend point at 1.24, and 15% above, in multiples of average earnings, with creditable earnings capped at 2.47. This schedule is Figure 2.

Under the Boskin proposal (equation 16) benefits are $b(x,j) = \max\{\underline{b},\, Ca_2/(1+g)^{j-J_r}\}$ for $j \geq J_r$, where C is a proportionality factor determined in equilibrium and $\underline{b}$ is the floor, set to 0, 0.15, 0.25 or 0.35 times output per person. The payroll tax rate θ and government consumption are held at their U.S.-system values throughout, as HV1999 p.514 requires.

Section 4 describes the third system the paper considers, in which the medical transfer is retained alongside the Boskin annuity.

3 Calibration and data inputs

The age-efficiency profile

κ_j, the age-efficiency profile, is taken from Hansen (1993). HV1999 footnote 16: "Hansen estimates median wage rates in cross-sectional data for males in different age groups. We use his values for the center of the age group and linearly interpolate to get the remaining values. We set the labor endowment in efficiency units of agents at a real-life age of 75 to zero." We follow this exactly, using Hansen's Table II male weights at the age-group centres $17.5, 22, 29.5, 39.5, 49.5, 59.5$ and running the profile linearly to zero at 75. The resulting profile is Figure 1, plotted with Hansen's tabulated values as markers.

Survival

Mortality is the U.S. male life table. The published table is given at five-year exact ages, so single-year values are obtained by interpolating $\log q_x$ rather than q_x — mortality is approximately Gompertz, i.e. log-linear in age, so linear interpolation of the level systematically understates mortality between knots. Survival at the terminal age is set to zero so that the cohort is exhausted at 100.

The two wage specifications

HV1999 considers two processes for log productivity, both with the dispersion parameter chosen so that the wage Gini for agents under 65 equals 0.35.

No idiosyncratic shocks. $(\gamma, \sigma_\epsilon^2) = (1, 0)$ and $\sigma_{y_1}^2 = 0.376$: ability is drawn at birth and perfectly preserved. This is a *permanent type* in the toolkit, not an exogenous state, and is implemented as

21 permanent types with masses from the binned normal.

Idiosyncratic shocks. $\sigma^2_{y_1} = 0.27$ with an AR(1), $z' = \gamma z + \epsilon$, $\epsilon \sim N(0, \sigma^2_\epsilon)$. HV1999 fix $\sigma^2_\epsilon = 0.015$ and $\gamma = 0.985$, chosen by the indirect-inference step described on p.512. Here z is a genuine exogenous state, and because $\bar{e}'$ and a'_2 depend on it the model uses `experienceassetz`.

The discretization follows footnote 17 exactly: "The shock z in each model takes on 21 evenly spaced values between $-4\sigma_{y_1}$ and $4\sigma_{y_1}$. Transition probabilities are calculated by integrating the area under the normal distribution conditional on the value of z." We use 21 points on that interval with transitions obtained by integrating the normal between grid midpoints, open-ended at both ends. Section 6 notes that this grid is coarse relative to the innovation standard deviation and that refining it moves the wage Gini; we keep 21 points because that is the paper's own choice.

4 Equilibrium and the financing of the floor

The stationary equilibrium conditions are: factor prices equal marginal products, the government budget balances, social security balances, and accidental bequests are absorbed by the government. Under the U.S. system the unknowns are r, w, τ, θ, b and the normalisation $\bar{y}^{\,\text{mean}}$, with θ set by social security balance and $b = 0.1242\,Y$.

The Boskin stage needs more care, because HV1999 specifies the financing of the floor in two separate places that must be read together. On p.514 the paper says "the proportionality factor C is then set so that benefit payments to current retirees equal current social security tax payments (condition 6 in the definition of equilibrium)". Taken literally and applied to *total* benefits, this would pin $\sum \max\{\underline{b}, Ca_2/(1+g)^{j-J_r}\} = \theta w L$, which drives C towards zero as the floor rises and leaves no residual to finance. But footnote 23 states that "transfers needed to finance $\underline{b} = 0.35Y$ are on the order of 1.3–1.4% of GDP in the no idiosyncratic case. This results in income tax rates going from 18.7% in the zero floor situation to 22.3% for $\underline{b} = 0.35Y$." Those two statements are consistent only if the payroll tax finances the *annuity* and the floor top-up is a transfer out of general revenue. We therefore impose

$$\begin{aligned} \theta\, w\, L &= \textstyle\sum_{j \geq J_r} Ca_2/(1+g)^{j-J_r}, \\ \tau(rK + wL) + \text{AccBeq} &= \tfrac{G}{Y}\, Y + \mathcal{T}, \\ \mathcal{T} &\equiv \textstyle\sum_{j \geq J_r} \Big[\max\big\{\underline{b},\, Ca_2/(1+g)^{j-J_r}\big\} - Ca_2/(1+g)^{j-J_r} \Big]. \end{aligned}$$

The first condition makes the annuity self-financing on a pay-as-you-go basis, which is the paper's "the financing of benefits is always on a pay-as-you-go basis" (p.515); the second is the government budget. The residual $\mathcal{T}$ — the sum of $\max\{0, \underline{b} - \text{annuity}\}$ over retirees — is the transfer footnote 23 refers to. At $\underline{b} = 0$ the bracket is identically zero and the system collapses to the zero-floor case, which is a useful internal check: the $\underline{b} = 0$ equilibrium computed under this formulation reproduces the one computed before it was introduced to every printed digit.

The reading is confirmed quantitatively. Footnote 23 is a statement about the no-idiosyncratic economy, so it is directly comparable with Table III: we obtain an income tax rate of **18.7%** at a zero floor — exactly the paper's figure — rising to 21.4% at $\underline{b} = 0.35Y$ against the paper's 22.3%, with a floor transfer of 1.82% of GDP against the paper's 1.3–1.4%.

The floor itself is a multiple of output per person and output is endogenous, so $\underline{b}$ is carried as an equilibrium unknown with the condition $\underline{b} = \text{frac} \times Y$, exactly as $b = 0.1242\,Y$ is handled in the

U.S. stage.

The Boskin proposal with medical benefits retained

HV1999 Section 4.2 observes that under the plain Boskin proposal median-ability agents lose the common medical transfer as well as the redistribution built into the concave PIA formula, and asks what happens if only the old-age component is reformed. The resulting third system is specified in footnote 29: the payroll tax splits as $\theta = \theta_1 + \theta_2$, where θ_1 is the rate that financed the common transfer in the U.S. economy and is held fixed, θ_2 becomes the rate at which earnings accumulate into a_2, and benefits at retirement are $\max\{\underline{b},\, b + b(a_2, j)\}$. We recover $\theta_1 = b\,\mu_{\text{ret}}/(wL)$ from the U.S. equilibrium, where μ_{ret} is the mass of retirees, and accumulate a_2 at $\theta_2 = \theta - \theta_1$. Agents continue to pay the full θ; the split governs only what each part finances.

One ambiguity should be recorded. Footnote 29's $\max\{\underline{b},\, b + b(a_2, j)\}$ is implemented here with the floor applying to the *total* retirement income from the system. The alternative reading, $b + \max\{\underline{b}, b(a_2, j)\}$, would apply the floor to the annuity alone and would make the floor bind for more retirees at a given $\underline{b}$.

Compensating variations

The welfare measure is the usual consumption-equivalent variation for a newborn of each ability level. Because period utility is $\left(c^{\nu}(1-l)^{1-\nu}\right)^{1-\sigma}/(1-\sigma)$, scaling an agent's consumption possibilities by $(1+x)$ at every date and state leaves optimal hours unchanged and scales lifetime utility by $(1+x)^{\nu(1-\sigma)}$. Hence

$$\text{CEV}_i = \left(\frac{V_i^{\text{Boskin}}(0,0,1)}{V_i^{\text{US}}(0,0,1)} \right)^{\frac{1}{\nu(1-\sigma)}} - 1,$$

evaluated at the newborn state (zero assets, zero $\bar{e}$ or a_2, age 1). This is exact for this utility function, so no re-solve or policy-evaluation step is required. The aggregate measures of Tables V and VI apply the same transformation to the ability-weighted expected newborn value, i.e. the welfare of an agent behind the veil of ignorance about ability.

5 Results

Aggregates: Tables III and IV

Tables 2 and 3 report the general features of the model economies under the two wage specifications. K/Y is within 0.05 of HV1999 in every row, and the *changes* across floor levels match closely: our K/Y falls by 0.13 across the no-idiosyncratic rows against the paper's 0.15, and by 0.12 across the idiosyncratic rows against the paper's 0.16. The interest rate tracks the paper's rise with the floor to within 0.2 percentage points and is exact at the two highest floors of Table IV. The income Gini matches at the two digits reported, in both specifications and at all four floors.

The share of retirees at the floor is close at the higher floors — 61.8 against 59.2 and 60.5 against 58.8 at $\underline{b} = 0.35Y$ — but too high at $\underline{b} = 0.15Y$, where we obtain 28.0 against 15.6 in Table III and

24.0 against 12.7 in Table IV. This is the one column where a middle row is materially worse than the rows either side of it, and it is discussed in Section 6.

Welfare: Tables V and VI, and Figures 3 and 4

The compensating variations are Figures 3 and 4. The qualitative result the paper is built on replicates. At a zero floor the profile is monotone in ability: the lowest-ability agents lose heavily (about −18% in the idiosyncratic economy) while the highest gain about +10%. At every positive floor the profile becomes U-shaped, and at $\underline{b} = 0.35Y$ the lowest-ability agents *gain* (+1.3%) while agents of median ability lose (−1.7%) — precisely HV1999 p.516, "low-ability and high-ability agents experience a welfare gain, but agents with median ability levels always suffer a welfare loss". The trough migrates towards higher ability as the floor rises, from ability level 5 at $\underline{b} = 0.15Y$ to level 11 at $\underline{b} = 0.35Y$.

Tables 4 and 5 give the aggregate welfare measures. The paper's two qualitative claims hold in our results: the ex-ante measure is negative at every floor level considered, and a majority of agents are worse off at birth under every version of the proposal. Retaining the medical transfer roughly halves the aggregate welfare loss — from −4.60 to −1.89 at a zero floor in the no-idiosyncratic economy, against the paper's −3.5 to −1.30 — which is the point of the exercise in Section 4. The levels, however, are about twice as negative as HV1999's throughout, and our share of agents worse off is roughly ten percentage points higher; see Section 6.

Figure 4 also reproduces the paper's observation that the compensating variations with medical benefits retained are "qualitatively similar" to those without: the same U-shape, over a much compressed range (−5 to +7 percent rather than −18 to +10). With the transfer retained the floor binds for far fewer retirees at low floor levels — 3.1% at $\underline{b} = 0.15Y$ against 24.0% without it — because b plus the annuity already clears the floor, which is why the $\underline{b} = 0$ and $\underline{b} = 0.15Y$ curves nearly coincide.

Life-cycle profiles: Figures 5 and 6

Figure 5 shows the mechanism behind the decline in aggregate labour input as the floor rises. Panel (c) isolates a low-ability agent (ability level 5): under the Boskin proposal with a zero floor this agent works *more* than under the U.S. system, but at $\underline{b} = 0.35Y$ hours fall below the U.S. profile from roughly age 40 onwards. This is HV1999 p.516: "low ability agents who will receive the floor benefit with certainty get no marginal benefit for an additional unit of social security taxes paid". Figure 6 shows the corresponding consumption profiles; in panel (c) the ability-5 profile jumps at age 65, where the floor benefit begins, a discontinuity absent at a zero floor.

6 Remaining differences

Labour input and hours. Aggregate L sits about 8% and the fraction of time worked about 11% above HV1999 in *every* row of both tables. The offset is a level effect, not a specification error: the paper's own L moves from 0.396 under the U.S. system to 0.413 under Boskin with a zero floor, and ours moves $0.427 \to 0.446$ — the same increment. The same holds across floor levels and across the two wage specifications. The most likely source is the normalisation of the Hansen

profile, which the paper does not state: any constant rescaling of κ_j shifts the level of L without affecting its response to policy.

The wage Gini and the dispersion of ability. The under-65 wage Gini is a calibration target of 0.35 in both specifications. The no-idiosyncratic economy hits it (0.352), but the idiosyncratic economy overshoots at 0.370. The statistic is exactly computable offline, since the wage is $w \exp(z)\kappa_j$ and a Gini is scale invariant, so w drops out and only the joint distribution of (z, age) matters. Doing so shows the overshoot is a discretization effect: at $n_z = 21$ the grid spacing on $[-4\sigma_{y_1}, 4\sigma_{y_1}]$ is 0.208 in logs while the innovation standard deviation is $\sqrt{0.015} = 0.122$, so Tauchen is being asked to represent shocks smaller than its own grid spacing. Refining to $n_z = 31$ and 51 gives Ginis of 0.359 and 0.353, extrapolating to about 0.345. We nonetheless retain $n_z = 21$, because that is what footnote 17 specifies; refining the grid would depart from the published method.

The share at the floor at $\underline{b} = 0.15Y$. At the middle-low floor we place roughly twice as many retirees at the floor as HV1999 do, while matching well at $0.25Y$ and $0.35Y$. This is the same direction as the wage-Gini overshoot: a wider distribution of accumulated contributions puts more retirees below a given floor. It is most visible at $0.15Y$ because that is where the floor cuts through the densest part of the annuity distribution, so a given error in dispersion translates into the largest error in the share.

The welfare aggregates. Tables V and VI are about twice as negative as the paper's and our share of agents worse off is about ten percentage points higher. Both follow from the same wider ability distribution: the share worse off is the mass below the CEV's zero crossing, and the ex-ante measure is an ability-weighted average of a profile that is steeply graded in ability. We also do not reproduce two features of the paper's tables — the non-monotone uptick in Table V at $\underline{b} = 0.35Y$ (the paper's measure worsens from -0.9 to -1.5 in the no-idiosyncratic column where ours improves monotonically), nor the single positive entry in either table, $+0.16$ at $\underline{b} = 0.25Y$ in the idiosyncratic column of Table VI, where we obtain -0.90.

Unstated parameters. Two quantities that matter are not reported in the paper: the upper bound on the asset grid, and the normalisation of κ_j. The appendix gives only "an upper bound imposed on the asset space". Our choices are documented in Appendix A.

TABLE III
General Features—No Idiosyncratic Shocks

	K/Y	L	Fraction of time working	r	Income Gini	Percentage of retired agents at floor level
U.S. system	2.91	0.396	0.314	6.4	0.40	—
Boskin proposal						
$\underline{b} = 0.0Y$	2.96	0.411	0.330	6.2	0.39	0.0
$\underline{b} = 0.15Y$	2.94	0.407	0.324	6.2	0.40	15.6
$\underline{b} = 0.25Y$	2.89	0.401	0.315	6.5	0.40	41.0
$\underline{b} = 0.35Y$	2.81	0.391	0.307	6.8	0.41	59.2

	K/Y	L	Fraction of time working	r	Income Gini	Percentage of retired agents at floor level
U.S. system	2.89	0.427	0.350	6.5	0.40	—
Boskin proposal						
$\underline{b} = 0Y$	2.92	0.445	0.370	6.3	0.39	0.0
$\underline{b} = 0.15Y$	2.89	0.439	0.360	6.4	0.40	28.0
$\underline{b} = 0.25Y$	2.85	0.430	0.350	6.6	0.40	47.3
$\underline{b} = 0.35Y$	2.79	0.420	0.342	6.9	0.40	61.8

Table 2: Table III of Huggett & Ventura (1999): general features, no idiosyncratic shocks. Top: original from paper. Bottom: replicated.

TABLE IV
General Features—Idiosyncratic Shocks

	K/Y	L	Fraction of time working	r	Income Gini	Percentage of retired agents at floor level
U.S. system	3.06	0.413	0.312	5.8	0.43	—
Boskin proposal						
$\underline{b} = 0.0Y$	3.11	0.434	0.329	5.6	0.43	0.0
$\underline{b} = 0.15Y$	3.08	0.429	0.322	5.7	0.43	12.7
$\underline{b} = 0.25Y$	3.02	0.421	0.312	6.0	0.44	39.1
$\underline{b} = 0.35Y$	2.95	0.413	0.306	6.2	0.44	58.8

	K/Y	L	Fraction of time working	r	Income Gini	Percentage of retired agents at floor level
U.S. system	3.04	0.446	0.346	5.8	0.43	—
Boskin proposal						
$\underline{b} = 0.00Y$	3.07	0.468	0.368	5.7	0.43	0.0
$\underline{b} = 0.15Y$	3.05	0.461	0.358	5.8	0.43	24.0
$\underline{b} = 0.25Y$	3.01	0.451	0.347	6.0	0.44	45.1
$\underline{b} = 0.35Y$	2.95	0.441	0.339	6.2	0.43	60.5

Table 3: Table IV of Huggett & Ventura (1999): general features, idiosyncratic shocks. Top: original from paper. Bottom: replicated.

TABLE V
Aggregate Welfare Gains and Agents with Welfare Losses at Birth

Model economy	No idiosyncratic shocks		Idiosyncratic shocks	
	Welfare gains (%)	Agents w/ losses (%)	Welfare gains (%)	Agents w/ losses (%)
$\underline{b} = 0.0Y$	−3.5	57.9	−2.4	72.5
$\underline{b} = 0.15Y$	−1.9	72.5	−1.3	72.5
$\underline{b} = 0.25Y$	−0.9	80.5	−0.3	72.1
$\underline{b} = 0.35Y$	−1.5	80.5	−0.9	88.3

Model economy	No idiosyncratic shocks		Idiosyncratic shocks	
	Welfare gains %	Agents w/ losses %	Welfare gains %	Agents w/ losses %
$\underline{b} = 0.00Y$	−4.60	72.6	−4.72	84.1
$\underline{b} = 0.15Y$	−3.19	84.1	−3.19	84.1
$\underline{b} = 0.25Y$	−2.00	82.7	−1.84	91.9
$\underline{b} = 0.35Y$	−1.41	80.5	−1.17	95.0

Table 4: Table V of Huggett & Ventura (1999): aggregate welfare gains and agents with welfare losses at birth, under the Boskin proposal. Top: original from paper. Bottom: replicated.

TABLE VI
Aggregate Welfare Gains and Agents with Welfare Losses at Birth
(Medical and Hospital Benefits Held Constant)

Model economy	No idiosyncratic shocks		Idiosyncratic shocks	
	Welfare gains (%)	Agents w/ losses (%)	Welfare gains (%)	Agents w/ losses (%)
$\underline{b} = 0.0Y$	−1.3	72.5	−0.80	72.6
$\underline{b} = 0.15Y$	−1.3	72.6	−0.81	72.6
$\underline{b} = 0.25Y$	−0.3	76.1	0.16	64.5
$\underline{b} = 0.35Y$	−0.4	69.0	−0.15	69.0

Model economy	No idiosyncratic shocks		Idiosyncratic shocks	
	Welfare gains %	Agents w/ losses %	Welfare gains %	Agents w/ losses %
$\underline{b} = 0.00Y$	−1.89	84.1	−1.79	84.1
$\underline{b} = 0.15Y$	−1.83	84.1	−1.75	84.1
$\underline{b} = 0.25Y$	−0.97	88.3	−0.89	91.8
$\underline{b} = 0.35Y$	−0.36	69.0	−0.23	69.0

Table 5: Table VI of Huggett & Ventura (1999): the same welfare measures with the medical and hospital transfer held constant across systems. Top: original from paper. Bottom: replicated.

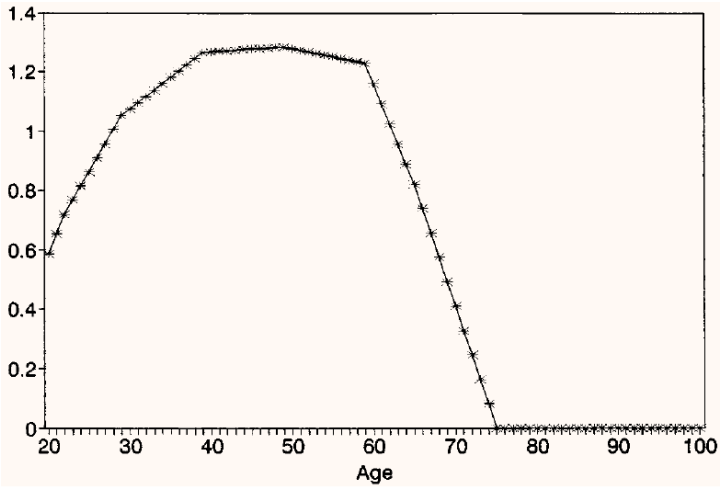


FIG. 1. Wage profile. Source: Hansen [8].

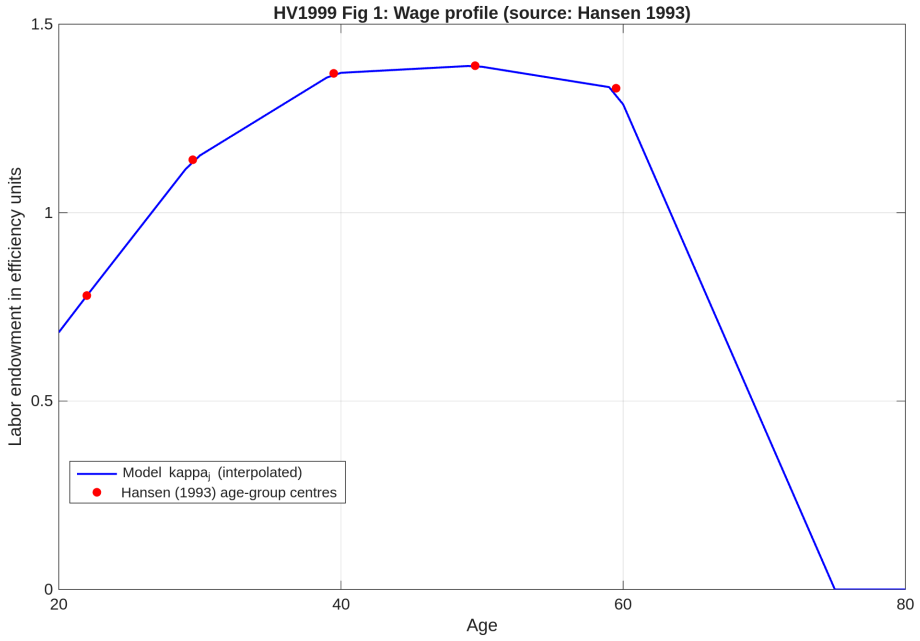


Figure 1: Figure 1 of Huggett & Ventura (1999): the age-efficiency profile. Top: original from paper. Bottom: replicated, with Hansen's tabulated age-group values shown as markers.

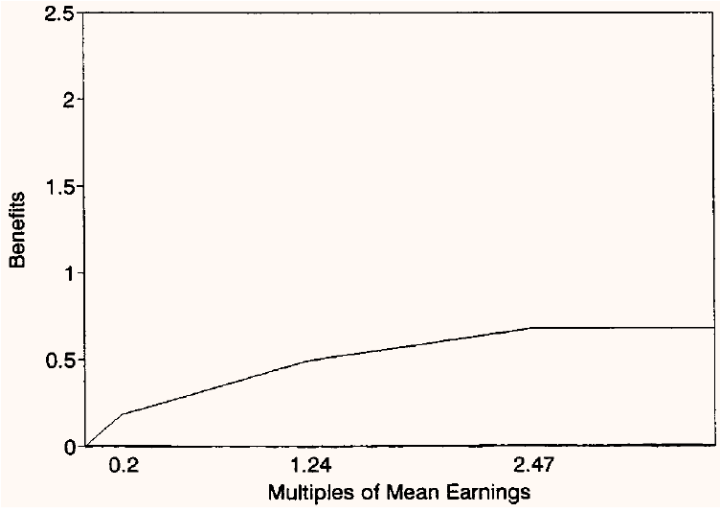


FIG. 2. Social security benefits and earnings-related components.

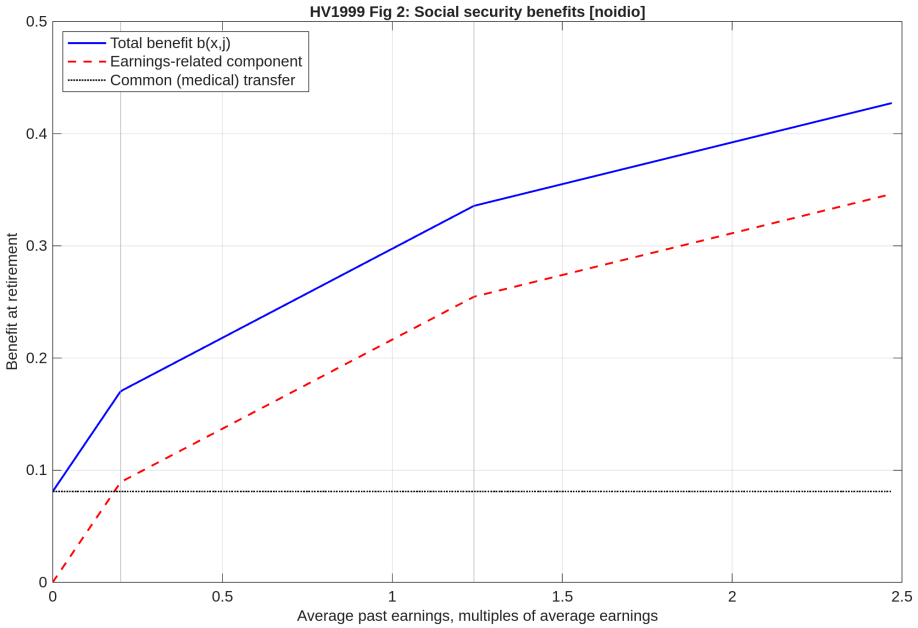


Figure 2: Figure 2 of Huggett & Ventura (1999): social security benefits as a function of average past earnings. Top: original from paper. Bottom: replicated, showing the total benefit, the concave earnings-related component, and the common medical transfer separately. The kinks are the bend points at 0.20 and 1.24.

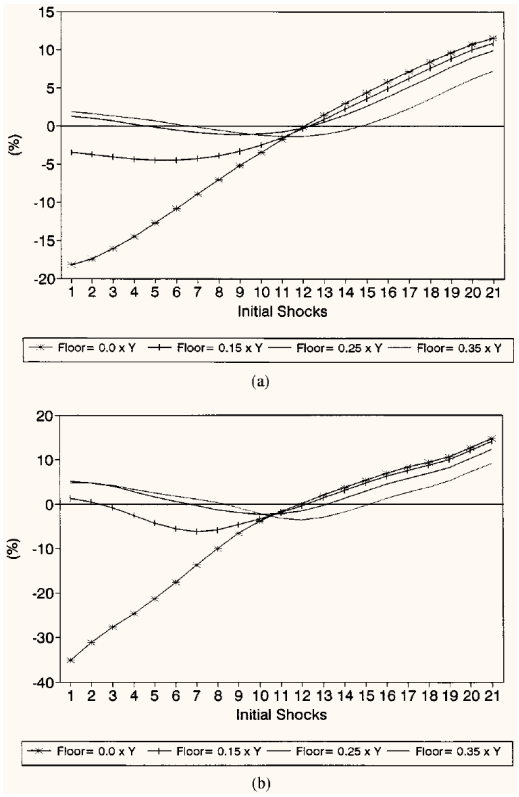


FIG. 3. (a) Compensating variations. Idiosyncratic uncertainty. U.S. system vs. Boskin proposal. (b) Compensating variations. No idiosyncratic uncertainty. U.S. system vs. Boskin proposal.

HV1999 Fig3: Compensating variations. U.S. system vs. Boskin proposal

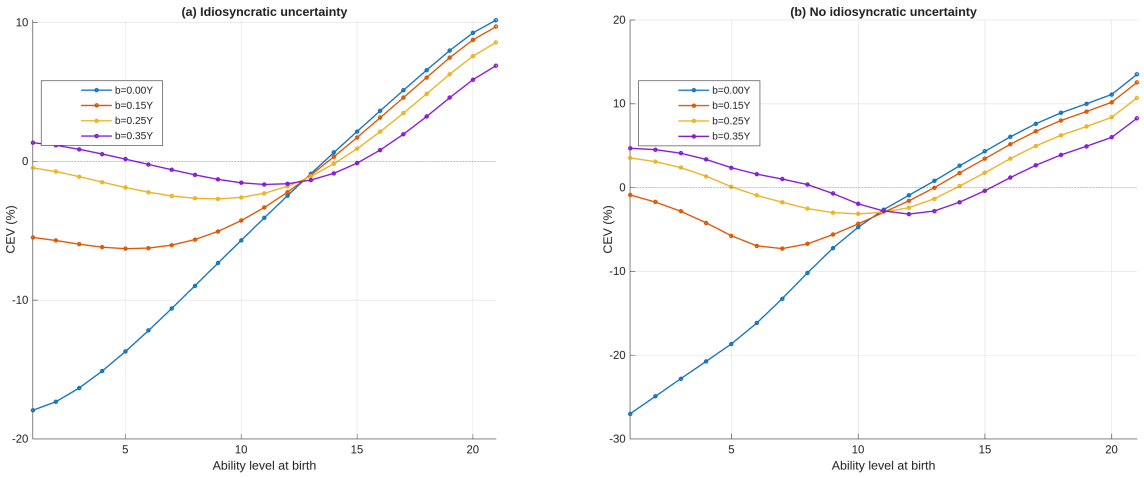


Figure 3: Figure 3 of Huggett & Ventura (1999): compensating variations, U.S. system versus Boskin proposal, by ability at birth. Top: original from paper, panel (a) with idiosyncratic uncertainty and panel (b) without. Bottom: replicated, same panel ordering.

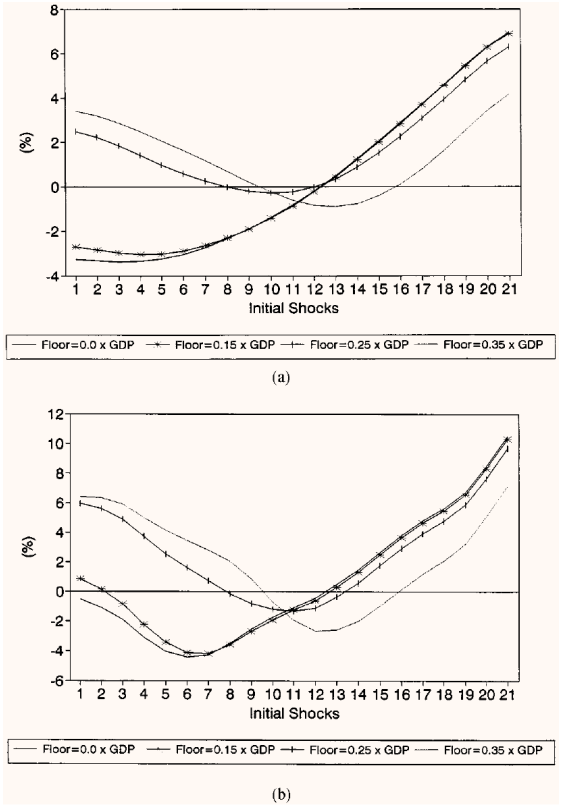


FIG. 4. (a) Compensating variations. Idiosyncratic uncertainty. U.S. system vs. Boskin proposal (w/medical benefits). (b) Compensating variations. No idiosyncratic uncertainty. U.S. system vs. Boskin proposal (w/medical benefits).

HV1999 Fig4: Compensating variations. U.S. system vs. Boskin proposal (w/medical benefits)

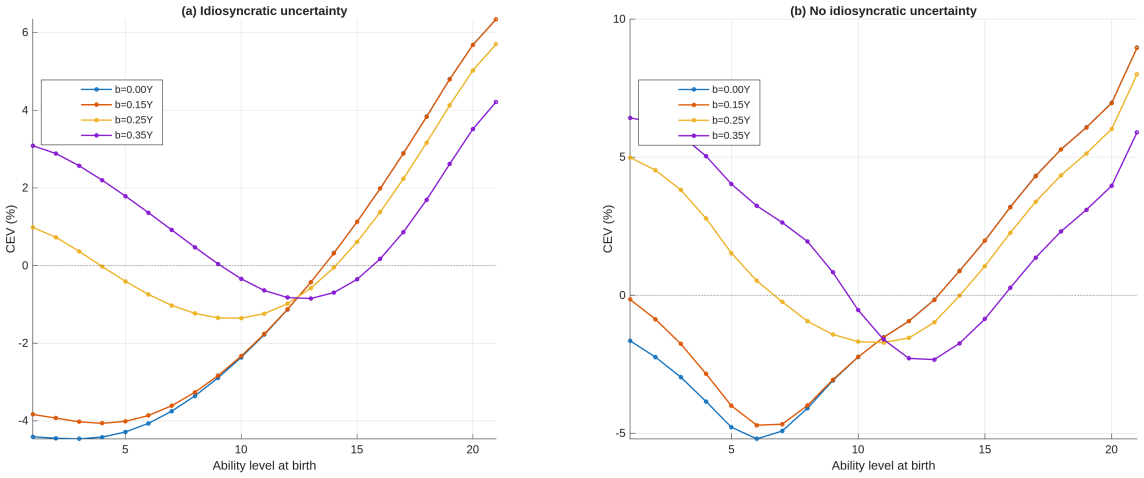
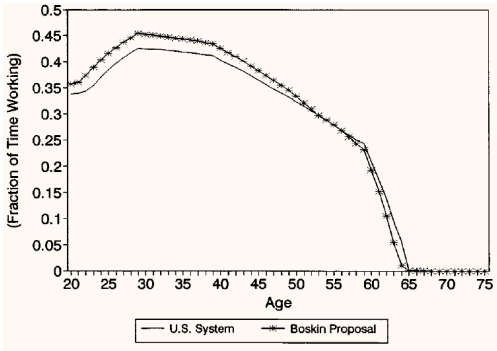
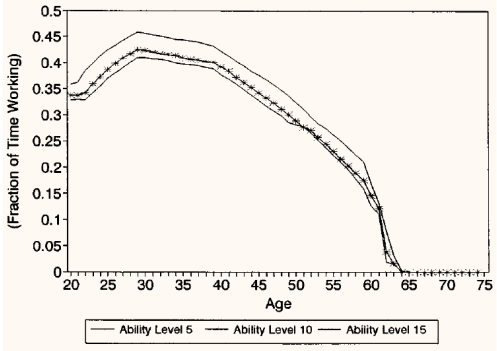


Figure 4: Figure 4 of Huggett & Ventura (1999): compensating variations with the medical and hospital transfer retained in both systems. Top: original from paper. Bottom: replicated.

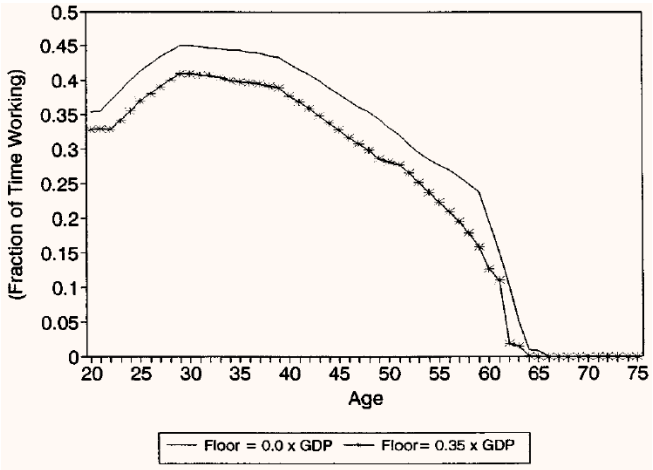


(a)



(b)

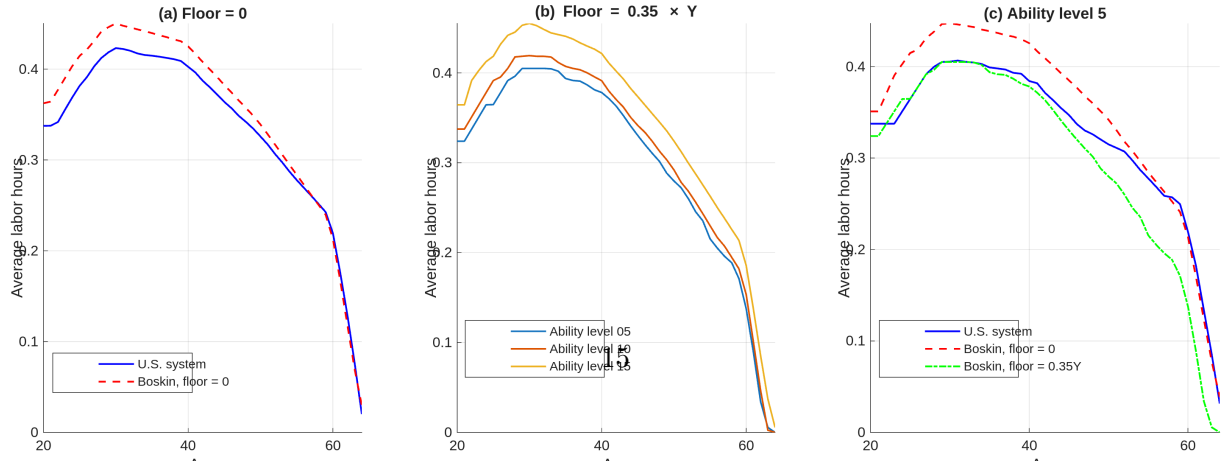
FIG. 5. (a) Average labor hours: No idiosyncratic uncertainty. U.S. system vs. Boskin proposal (floor = 0). (b) Labor hours profile. No idiosyncratic uncertainty (floor = $0.35 \times Y$). (c) Labor hours of low-ability agents. No idiosyncratic uncertainty (ability level 5).



(c)

FIGURE 5—*Continued*

HV1999 Fig 5: Labor hours. No idiosyncratic uncertainty



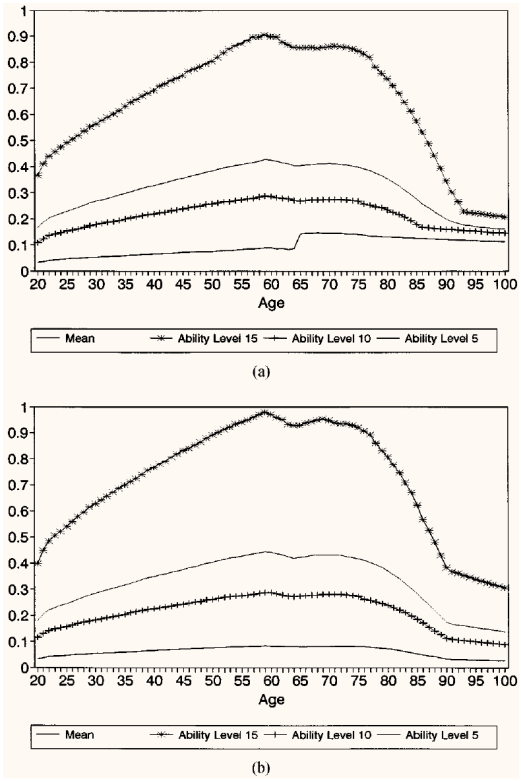


FIG. 6. (a) Consumption profile (U.S. system). No idiosyncratic uncertainty. (b) Consumption profile (Boskin proposal). Zero floor; no idiosyncratic uncertainty. (c) Consumption profile (Boskin proposal). Floor = 0.35 × GDP; no idiosyncratic uncertainty.

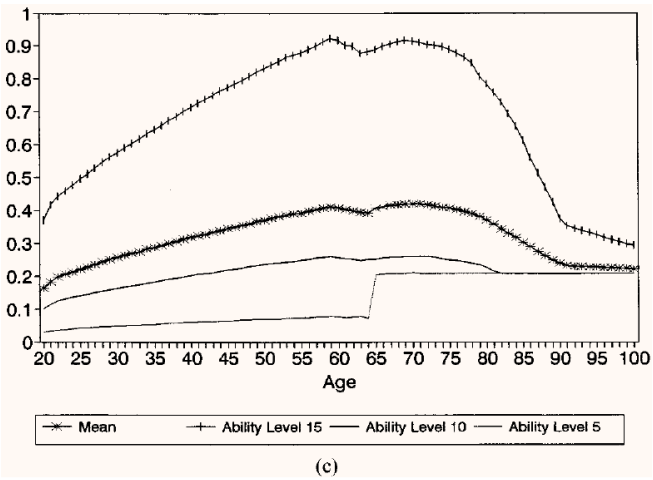
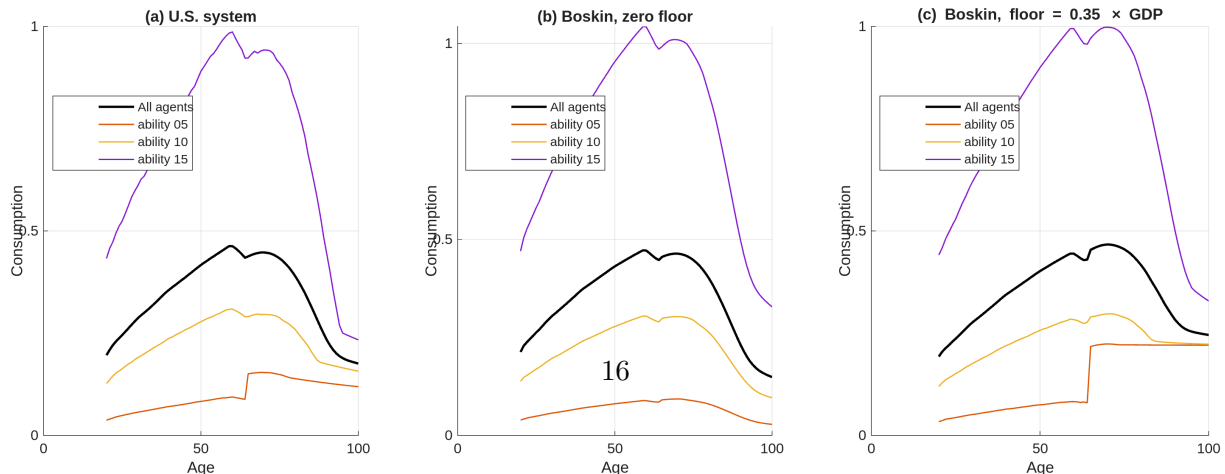


FIGURE 6—*Continued*

HV1999 Fig 6: Consumption profiles. No idiosyncratic uncertainty



A Computational details

State space. The U.S. system has endogenous states $(a, \bar{e})$ on a 251×25 grid and the Boskin systems (a_1, a_2) on 251×50. The asset grid uses the paper's own curvature, $a_k = \bar{a}(k/K)^{2.35}$, with $\bar{a} = 40$; the private asset grid is identical across systems, because the compensating variations compare the two steady states at the same newborn state. HV1999 uses 101 points on assets, 4 on $\bar{e}$ and 21 on z; we keep 21 on z (footnote 17) but refine the endogenous grids, because four points cannot carry an accumulating average. The labour choice is discretized at 75 points, where the paper uses golden-section search; moving between 51 and 101 points changes every aggregate by less than one part in 10^4, and leaves the compensating variations unchanged at the printed precision.

The a_2 ceiling. Accumulated contributions are bounded above in this model rather than merely unlikely to be large: the productivity grid tops out at $\exp(4\sigma_{y_1})$, so even an agent drawing the highest shock every year and working $l = 1$ every year reaches only $a_2 = 98.0$ at the equilibrium prices of the idiosyncratic economy. The ceiling is sensitive to r through 45 years of compounding at $(1 + r)/(1 + g)$ — 79.6 at $r = 5\%$, 133.0 at $r = 7\%$ — so the grid is taken to 150, which cannot bind for any interest rate below about 7.3%. Diagnostics confirm it: the mass in the top ten a_2 grid points is 7.6×10^{-10}.

Grid-top diagnostics. Every stage reports the mass sitting in the top grid points of each endogenous state. The a_2 ceiling is never approached. The asset ceiling $\bar{a} = 40$ carries 1.7×10^{-4} of the mass in its top ten points under the idiosyncratic U.S. system, slightly above our 10^{-5} reporting threshold; widening it moves an income Gini that currently matches the paper exactly, so it is reported rather than tuned away.

Solution method. Value function iteration with divide-and-conquer and a grid-interpolation layer on the asset dimension. General equilibrium is solved by Anderson acceleration of the shooting map, which converges in eight to ten iterations from a warm start at every floor level. Successive floor levels are solved in increasing order, each starting from the previous level's equilibrium.

Reproducing these results. The driver is `HuggettVentura1999.m`, with the wage specification selected by `idioShocks` and the work split into seven stages by the `doPart` vector: the U.S. steady state; the Boskin proposal at each floor level; the compensating variations; the medical-benefits variant at each floor level; its compensating variations; the age profiles; and the figures. The floor levels solved in a given job are selected by `doFloorsB` and `doFloorsM`, so a single level can be re-run on its own, and each level writes its own `.mat`. Tables III–VI are assembled in the final stage from those files, so levels solved in different jobs combine.